

Isocolon, parison, and comparable-set construal: A corpus test of formal balance

Brett Reynolds *

Humber Polytechnic & University of Toronto

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Abstract

Rhetorical theory often treats balanced form as more than ornament. A balanced pair of cola can present two units as candidates for comparison. A reader-side causal claim would need experimental evidence, but a production-side textual claim can be tested in corpus data. This paper tests whether adjacent discourse-relation pairs in the Georgetown University Multilayer corpus with enhanced Rhetorical Structure Theory annotations (GUM/eRST) show more **FORMAL BALANCE** when they construe two units as comparable co-members of a set. The analysis separates **ISOCOLON**, measured as length balance, from **PARISON**, measured as syntactic parallelism, and **LEXICAL ECHO**. In 17,456 adjacent relation pairs from 299 documents across 24 genres, a broad comparison and co-ordination family shows a positive adjusted increase in composite formal balance. The largest broad-family effects come from lists and disjunctions, where co-ordination is a strong grammatical null; the cleaner non-enumerative signal comes from adversative contrast.

The composite increase comes mostly from parison. Mean-only length controls also show length balance, but they contribute little to the composite signal. A matched co-ordination check shrinks the list/disjunction effect; an adversative-only check leaves contrast positive. Antithesis alone sits near zero, and concession is negative. The result supports a restrained textual claim: formal balance belongs to the profile of some comparable-set construals and varies across relations that could be called contrastive.

I INTRODUCTION

Rhetorical theory has long treated figures as patterned forms with argumentative force. In that tradition, **ISOCOLON** joins two kinds of evidence that are easy to confuse: formal balance between adjacent units and a reader's sense that the units invite comparison. The motivating claim is simple. When writers place two clauses or phrases in balanced form, they may present the comparison between them as more explicit, pointed, or forceful.

This paper asks how much of that claim can be tested without a reader experiment. The answer depends on the explanatory level. At the reader-response level, the claim says that isocolon changes

*Contact: brett.reynolds@humber.ca

what readers notice or how forcefully they construe a relation. Corpus evidence alone can't show that. At the textual level, the claim turns distributional: writers may preferentially use balanced form where they construe two units as comparable. That claim is testable.

The corpus test uses a narrow hypothesis. When a relation places two adjacent units in a shared comparison frame, the pair should show more formal balance than pairs expressing cause, condition, elaboration, sequence, or other non-comparative relations. The study uses the Georgetown University Multilayer (GUM) corpus and enhanced Rhetorical Structure Theory (eRST) annotations because they provide genre-diverse texts, discourse-relation labels, and signal annotation (Zeldes, 2017; Zeldes et al., 2025).

The analysis uses the eRST relation export: mostly GUM, plus a smaller GENTLE challenge set included in the same export. Access to the Rhetorical Structure Theory (RST) Signalling Corpus is still pending, so that corpus isn't part of this analysis. Once available, it can support later triangulation against a broader inventory of coherence-relation signals.

The pattern refines the rhetorical claim. Broadly comparative and co-ordinative relations are more formally balanced than matched non-target relations, but the largest broad-family effects occur exactly where a grammar-driven co-ordination account is strongest: lists and disjunctions. Ordinary adversative contrast is smaller but cleaner, because it survives inside an adversative-only comparison. Antithesis alone is near zero, and concession is negative. The pattern suggests that formal balance, especially parison-like syntactic parallelism, tracks some acts of comparative construal better than it tracks the inherited relation name ANTITHESIS.

2 RHETORICAL FIGURES AS FORM-FUNCTION PAIRINGS

The analysis starts from Fahnestock's account of rhetorical figures as form-function pairings. In her work on scientific argument and rhetorical style, figures are recurrent formal arrangements that can epitomize lines of reasoning (Fahnestock, 1999, 2011). That view gives the corpus prediction. If a figure serves an argumentative operation, then the figure should be distributed unevenly across the discourse contexts where that operation is being performed.

Fahnestock's discussion of the figures of argument speaks directly to this project. ISOCOLON concerns approximate equality in the length of adjacent cola, while PARISON concerns similarity of grammatical structure.

The two often co-occur, but the properties differ. A pair can be close in length without being syntactically parallel, and it can be syntactically parallel without being closely balanced in length. The classical tradition links both to comparison and contrast because similar form can make two units cohere as an argumentative set (Fahnestock, 2011).¹

A figure can help make a relation salient without being obligatory whenever the relation is present. For the corpus test, uneven distribution is expected, and frequency by itself gives no direct measure of the rhetorical value of the figure.

The same point motivates computational work on rhetorical figures. Harris and Di Marco (2017) argue that figural form is often iconically related to discourse function, so surface patterning can provide evidence about pragmatic and argumentative actions. Harris et al. (2018) develop an annotation scheme for figures in part because figure identification requires attention to both formal

¹I approach these terms as a corpus linguist with recent rhetorical reading. That outsider position matters: the Greek terms motivate hypotheses, but the corpus tests the mappings. Less formally, this study stays suspicious of its own Greek.

regularity and rhetorical role. Lawrence et al. (2017) make the parallel point for argument mining: rhetorical figures can identify regions where argumentative work is being done, but only if formal detection is connected to the content and relation being advanced.

Together, these studies motivate the working assumption. The analysis pairs surface balance with a discourse relation that makes two units available for comparison. The operational question asks whether formal balance, separated into isocolonic length balance, parison-like syntactic parallelism, and lexical echo, is higher in the relations where rhetorical theory predicts that balance has value.

3 FROM CANDIDATE FORM TO RHETORICAL FUNCTION

The primary methodological challenge concerns over-attribution. Repetition and parallelism are common in ordinary text, and only some instances have the rhetorical force that figure theory cares about. Work on automatic chiasmus detection makes the problem vivid. Dubremetz and Nivre (2018) show that trivial repetition patterns are easy to find, but repetitions with rhetorical effect are much harder to separate from accidental or irrelevant patterns. They treat figure detection as a ranking problem instead of a clean binary classification problem. The problem is not finding parallelism. The problem is persuading it to mean something.

That lesson carries over to isocolon and parison. Exact repetition, list formatting, dialogue formulas, quoted slogans, and segmentation artifacts can all look formally balanced. Some are rhetorically relevant, but many aren't instances of the target phenomenon. Recent survey work makes the broader point: rhetorical figure detection remains limited by small datasets, inconsistent definitions, and over-reliance on rule-based surface cues (Kühn & Mitrović, 2024; Kühn et al., 2024). A credible corpus test should use surface scores cautiously, validate them against human judgment, and avoid treating high balance as equivalent to a gold figure label.

The analysis below follows that strategy. The composite formal-balance score is a continuous outcome in the distributional analysis and a ranking score in validation. It ranks adjacent relation pairs by how much balance they display. The qualitative audit then asks whether high-scoring examples are plausible rhetorical-balance candidates after excluding exact repetition, dialogue formulas, and other false-positive types. This more modest target than automatic figure recognition is the right target for a first rhetorical-function test.

4 CORPUS AND MEASUREMENT

The corpus comes from the English Georgetown University Multilayer corpus with enhanced Rhetorical Structure Theory annotations (GUM/eRST). GUM provides genre-diverse, richly annotated English texts (Zeldes, 2017). eRST extends RST-style discourse annotation with graph structures and explicit signal annotation across spoken and written genres (Zeldes et al., 2025). The DISRPT shared-task export used here also includes a smaller GENTLE challenge set. The project needs discourse relations as well as adjacent sentences. It also needs enough genre variation to separate a rhetorical effect from a register-specific habit.

The analysis uses adjacent discourse-relation pairs. The dataset contains 17,456 such pairs from 299 documents across 24 genres. These discourse units serve as candidate cola-like spans, with no claim that they're classical cola in a strict prosodic sense.

A relation enters the analysis if eRST marks it as explicit or implicit, both spans have token intervals, and the two intervals are adjacent in document order. Non-adjacent linked spans are excluded.

Concurrent graph relations are kept as pairwise relation rows when their spans meet this adjacency criterion.

The DISRPT shared-task relation export used here omits the full eRST graph-node inventory from the derived table. The diagnostics in Table 1 are interval-level checks instead of a complete graph-theoretic audit. The extracted set contains 16,278 rows with collection label GUM and 1,178 with collection label *GENTLE*, the out-of-domain challenge set; the 24 genres are the document-genre labels in the extracted files.

Extraction diagnostic	Count
Valid explicit/implicit relation rows	34,976
Adjacent relation rows used in analysis	17,456
Non-adjacent relation rows excluded	17,520
gapped rows excluded	15,349
interleaved rows excluded	2,171
Adjacent rows with discontinuous token intervals	1,963
Exact duplicate adjacent span pairs	0
Adjacent explicit rows	4,943
Adjacent implicit rows	12,513

Table 1: Extraction diagnostics for the adjacent-pair analysis. Counts come from token intervals in the DISRPT shared-task relation export. Exact duplicate span pairs are counted by document, unit-1 tokens, and unit-2 tokens.

For each pair, I computed three surface measures. First, ISOCOLON measures length balance over token, character, and estimated syllable length. Second, PARISON measures similarity in dependency and lexical-category profiles. Third, LEXICAL ECHO captures repeated or closely corresponding lexical material. A composite score combines the three as a graded measure of formal balance. The score gives evidence about balance, not a figure label for the pair.

The component scoring stays deliberately simple. For any two units with positive length, a balance component is $1 - |n_1 - n_2| / \max(n_1, n_2)$. The length score averages this quantity over word tokens, characters, and estimated syllables. The syntax score averages normalized edit similarity over Universal Dependencies *UPOS* lexical-category tags, language-specific *XPOS* lexical-category tags, and dependency-relation sequences read in token order. I treat this as a surface syntactic profile, not a tree-alignment measure. The lexical score averages lemma-sequence edit similarity and Jaccard overlap over content lemmas.

The GUM/eRST files use the Conference on Natural Language Learning format for Universal Dependencies (CoNLL-U), which records tokenization, lemmas, lexical-category tags, and dependency labels. Punctuation tokens are excluded from the word, syntax, and lexical sequences, while colon and semicolon position are kept as separate controls. Syllable counts are approximate vowel-group counts.

The composite equals

$$0.40 \times \text{length} + 0.45 \times \text{syntax} + 0.15 \times \text{lexical}.$$

These weights define only the composite formal-balance score. The components are also reported separately, so the analysis can distinguish isocolonic length balance from parison-like syntactic parallelism.

The target relations were defined before the final robustness checks as analysis groups, separate from a new annotation layer. Table 2 gives the mapping. The broad target family includes *adversative-antithesis*, *adversative-contrast*, *joint-disjunction*, and *joint-list*. This family tests whether formal balance rises in relations that place units in a comparable or co-ordinated frame. The narrow target family includes only *adversative-antithesis* and *adversative-contrast*. Individual-relation analyses separate the broad effect into its component labels.

Analysis role	Labels, rationale, and caveat
Broad target	<i>adversative-antithesis</i> ; <i>adversative-contrast</i> ; <i>joint-disjunction</i> ; <i>joint-list</i> . Contrast, alternatives, and co-enumeration place units into a shared frame. The joint labels are co-ordination-rich and also the strongest grammar-driven null.
Narrow adversative target	<i>adversative-antithesis</i> ; <i>adversative-contrast</i> . Tests whether the broad result survives without list and disjunction labels.
Cleaner non-enumerative test	<i>adversative-contrast</i> . Tests contrast against other adversative relations, where list syntax is not the main explanation.
Contrastive control	<i>adversative-concession</i> . Adversative but often asymmetric; predicted to be weaker or negative.
Surface positive control	<i>restatement-repetition</i> . Expected to score high by form alone; used as a false-positive diagnostic, separate from the comparison target.

Table 2: Mapping from eRST relation labels to analysis roles. The table defines the operational grouping. Independent annotation of comparable-set construal remains future work.

The primary comparisons ask a plain difference question: after removing obvious confounds, do target relations receive higher balance scores than other relations? The controls remove differences that can make the comparison too easy: document, explicit signalling, sentence adjacency, punctuation, mean length, and maximum length. Document controls mean that, where possible, target and non-target pairs are compared inside the same text.

For readability, I report effects as score-point differences on a 0–100 version of the balance scale. A difference of 4.1 points means that target pairs score about four points higher than comparable non-target pairs after the controls. I estimate those differences with ordinary least squares, a regression method for average differences. The residualized version used here first removes the controls from both the target marker and the outcome, then compares what remains.

For the composite score, the controls matter because longer spans have more opportunities for structural mismatch. For the length subscore, the pair of length controls overcontrols the target quantity: token-length balance is $2\mu/m - 1$, where μ is mean span length and m is maximum span length. The isocolon subscore is reported with mean-length control and without maximum-length control.

The main controls are an intercept, explicit relation status, same-sentence adjacency, colon punctuation, semicolon punctuation, standardized log mean length, standardized log maximum length, and document fixed effects. Document-clustered intervals treat pairs from the same document as related observations rather than as wholly independent pieces of evidence.

A genre-varying analysis asks whether the effect belongs to only one or two genres. It first estimates the same adjusted effect within each genre, then uses a normal empirical-Bayes model. In plain terms, that model lets noisy genre estimates borrow strength from the overall pattern while still estimating how much genres vary around the shared mean.

Robustness checks add mean-only length models, joint-only and adversative-only comparisons, GUM-only and continuous-span-only subsets, leave-one-genre influence checks, score-weight sens-

itivity checks, and stratified permutation checks. In the permutation checks, the target labels are shuffled inside matched strata. The question is whether effects this large appear when relation labels are randomly assigned within comparable document, length, punctuation, and explicitness groups.

5 VALIDATION

The validation asks whether the score ranks useful cases near the top. It doesn't ask whether the score can label figures automatically. The eRST *syn-prl* signal labels mark cases where annotators identified syntactic parallelism as a discourse signal. Those labels give a small check on formal parallelism, not on rhetorical force.

I report the area under the receiver-operating characteristic curve (AUC) for its simple ranking interpretation. If one labelled *syn-prl* pair and one unlabelled pair are drawn at random, AUC is the chance that the score ranks the labelled pair higher. A value of 0.5 means chance ranking; a value of 1.0 means perfect ranking. The full receiver-operating characteristic curve plots recall against false-positive rate at every possible cutoff, but the ranking interpretation is the point needed here.

On that scale, the composite score has an AUC of 0.73, with an approximate Hanley–McNeil interval from 0.66 to 0.80. The parison subscore is slightly higher at 0.75 [0.69, 0.82], lexical echo is 0.72 [0.65, 0.79], and isocolonic length balance is 0.65 [0.58, 0.72] (Table 3). The interval gives a rough uncertainty range for the AUC, not a new effect size.

The threshold numbers say how useful the ranked list would be for audit. Precision asks how many of the top-scoring pairs carry a *syn-prl* label. Recall asks how many of all *syn-prl* pairs appear in that top slice. Lift compares the precision to the label's overall prevalence. At the 95th percentile, precision reaches only 3%, but recall reaches 37%, a lift of 7.4. At the 97.5th percentile, precision rises to 5.5%, recall reaches 33%, and lift reaches 13.1.

Those numbers look poor for automatic labelling but adequate for ranking rare candidates for audit. The validation also stays narrow: the *syn-prl* labels validate a surface parallelism signal. In this implementation, a pair is positive if either unit contains a token-level *syn-prl* signal in the eRST export, so the validation uses span-level signal presence as a proxy for relation-level syntactic parallelism. It doesn't validate rhetorical function itself.

Score	AUC	Gold mean	Non-gold mean
Formal balance	0.73 [0.66, 0.80]	0.50	0.33
Isocolon	0.65 [0.58, 0.72]	0.68	0.56
Parison	0.75 [0.69, 0.82]	0.44	0.21
Lexical echo	0.72 [0.65, 0.79]	0.20	0.06

Table 3: Score validation against the 73 eRST *syn-prl* signal labels in 17,456 adjacent relation pairs. AUC gives the chance that a random labelled pair receives a higher score than a random unlabelled pair. Brackets give approximate Hanley–McNeil 95% intervals for AUC.

The qualitative audit supplies a pilot rhetorical-function check. I coded 80 candidate pairs in a single-author, non-blind audit; row strata and scores were visible. Automated reviews only flagged borderline cases after coding. The audit isn't interrater-reliability evidence. It identifies false-positive classes and illustrative examples.

The pattern is the point: high broad-target candidates usually survived the audit, while repetition controls didn't. Exact repetition gives strong surface balance, so it functions as a control rather than a target.

The audit also clarified the boundary of the category. Bare identity, apposition, dialogue repair, and exact repetition can be rhetorically relevant apart from isocolon. Conversely, weak length balance can still matter when the two units share a frame strongly enough to invite slot substitution. The coding decisions treat isocolon as a component of a broader rhetorical-balance judgment, with false-positive classes instead of a mechanically recoverable surface pattern.

Sample stratum	<i>n</i>	Yes	No	Unc.	Main issue
High broad target	16	14	2	0	Slot substitution
Mid broad target	12	7	4	1	Weak frames
Narrow adversative	14	6	8	0	Label mismatch
High non-target/artifact	19	4	12	3	Dialogue/apposition
Concession candidates	5	1	4	0	Asymmetry
Repetition controls	9	0	9	0	Exact identity
Baseline non-target	5	1	4	0	Weak or no frame

Table 4: Human qualitative audit summary for 80 candidate pairs.

Type	Unit 1	Unit 2	Judgment
Slot substitution	<i>You can use this on twist outs.</i>	<i>You can use this on braid outs.</i>	Yes
Adversative contrast	<i>so that the broad sides of the cuttings face east and west</i>	<i>while the slim sides face north and south.</i>	Yes
Repetition	<i>Get your feet off of there.</i>	<i>Get your feet off of there.</i>	No
Label mismatch	<i>We are not setting out to turn them into programmers or data crunchers,</i>	<i>but want to boost their knowledge level</i>	No

Table 5: Illustrative audit examples. The final row is an *adversative-antithesis* relation label. It lacks the balanced opposed cola of classical antithesis.

6 RESULTS

The broad target family is more formally balanced than the comparison set. On a 0–100 version of the composite score, target pairs are 4.1 points higher after controls. The document-clustered interval runs from 3.6 to 4.7 points. That shift is modest but stable: about 0.28 standard deviations of the observed composite score.

The components show what kind of balance carries the effect. The parison difference is 7.7 points, compared with 2.4 points for lexical echo. With mean-length control only, the broad isocolon difference is 6.6 points. Table 6 reports the composite and the three components separately.

The broad-family result is an omnibus check. Its strongest labels are also the labels where co-ordination makes parallel form easiest, so the cleaner rhetorical test comes from more focused comparisons. The narrow adversative family, *adversative-antithesis* plus *adversative-contrast*, shows a smaller 1.3-point composite difference, with an interval from 0.5 to 2.1 points.

The individual labels explain the difference. *joint-disjunction* has an adjusted formal-balance effect of 4.8 points, *joint-list* has an adjusted effect of 4.4 points, and *adversative-contrast* has the smaller but more diagnostic effect of 2.4 points. *adversative-antithesis* is near zero at 0.4 points, with an interval that crosses zero. I read this as a mismatch between a discourse-relation label and the classical

figure. It gives no evidence against antithesis as a rhetorical resource. *adversative-concession* is negative at -1.3 points.

The relation-level scan also identifies an important positive control. *restatement-repetition* has the strongest adjusted composite effect, 19.0 points, with especially large lexical and parison effects. That’s exactly what a surface score should find: exact identity is highly balanced, with lexical echo stronger than the syntactic effect. The human audit shows why it needs separate treatment. Repetition can signal clarification, identity, repair, emphasis, or other rhetorical work; exact repetition remains distinct from balanced comparison. The analysis treats restatement/repetition as a control and diagnostic, separate from the comparison hypothesis.

Genre-varying estimates point in the same direction. The broad target has a partially pooled mean difference of 3.9 points across 20 genres. The *joint-list* estimate is 4.0 points across 17 genres, and *adversative-contrast* is 2.9 points across the genres where there’s enough evidence to estimate it. The narrow adversative estimate is 1.2 points. The between-genre standard deviation is about 1.1 points for the broad target, 1.3 for *joint-list*, 0.9 for the narrow adversative target, and 0.7 for *adversative-contrast*. The broad effect appears in multiple genres.

Target	Target <i>n</i>	Other <i>n</i>	Formal	Length	Isocolon	Parison	Lexical
Broad family	2,462	14,994	4.1	0.8	6.6	7.7	2.4
Narrow adversative	605	16,851	1.3	-0.1	4.7	2.4	1.7
<i>joint-list</i>	1,690	15,766	4.4	1.0	6.5	8.2	2.0
<i>joint-disjunction</i>	167	17,289	4.8	0.9	3.6	8.6	4.0
<i>adversative-contrast</i>	262	17,194	2.4	-0.0	7.3	4.5	2.8
<i>adversative-antithesis</i>	343	17,113	0.4	-0.2	2.5	0.8	0.9
<i>adversative-concession</i>	679	16,777	-1.3	0.1	1.1	-2.7	-0.7
<i>restatement-repetition</i>	321	17,135	19.0	4.6	17.0	27.3	32.5

Table 6: Selected adjusted score-point differences on composite formal balance and its components, using a 0–100 rescaling of the outcome. Target *n* gives the number of target rows; comparison *n* gives the number of rows outside that target. ISOCOLON is length balance, PARISON is syntactic parallelism, and LEXICAL ECHO is lemma-sequence similarity plus content-lemma overlap. Formal, length, parison, and lexical estimates use document fixed effects and controls for explicitness, sentence adjacency, punctuation, mean length, and maximum length. The ISOCOLON column uses the same length-balance outcome but omits maximum length because mean and maximum length nearly determine the length-balance score. For that reason, the formal composite is not the weighted sum of the displayed ISOCOLON, PARISON, and LEXICAL ECHO columns. The restatement row is a positive control, separate from the comparison target.

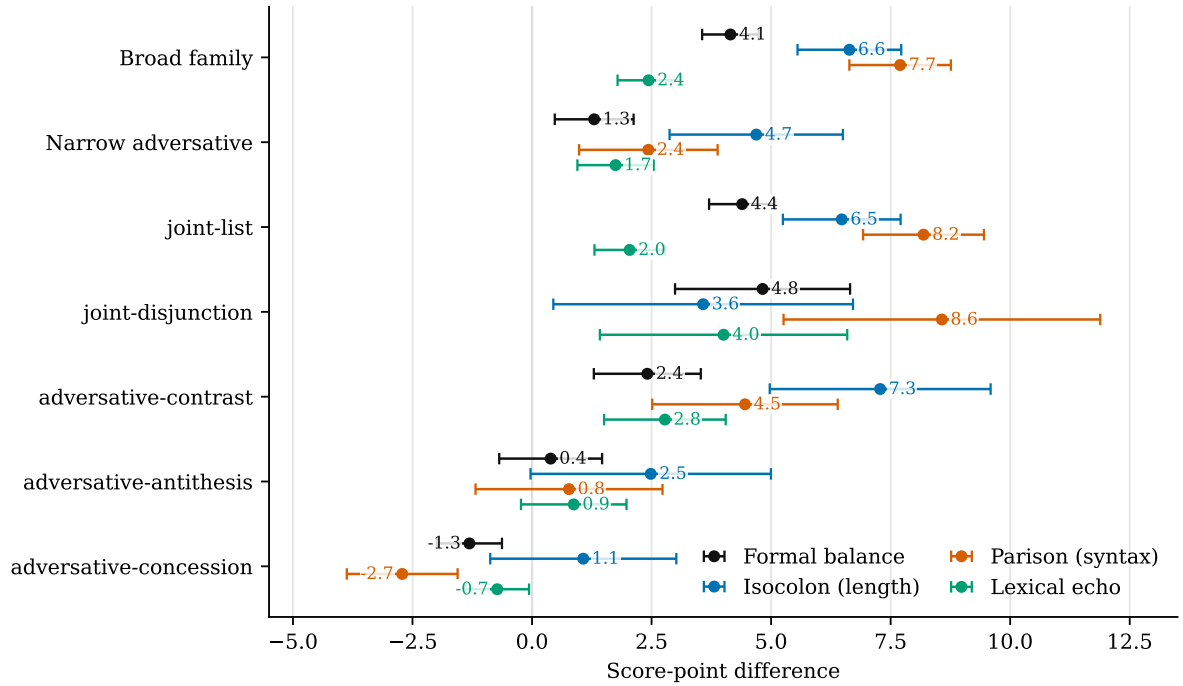


Figure 1: Adjusted score-point differences for FORMAL BALANCE, ISOCOLON, PARISON, and LEXICAL ECHO. Points are estimates; horizontal intervals are document-clustered 95% intervals. Effects are shown as score-point differences on a 0–100 scale. The ISOCOLON estimates use mean-length controls and omit maximum length; the other estimates use the main control specification.

6.1 ROBUSTNESS CHECKS

The primary formal-balance results survive document-clustered uncertainty estimates. With document clustering, the broad-family effect is 4.1 points, with an interval from 3.6 to 4.7. The *joint-list* effect is 4.4 [3.7, 5.1], *joint-disjunction* is 4.8 [3.0, 6.7], and *adversative-contrast* is 2.4 [1.3, 3.5]. *adversative-antithesis* remains near zero, 0.4 [−0.7, 1.5].

The length-balance result uses a different control specification. Mean span length and maximum span length nearly determine the length-balance score, so using both controls suppresses the quantity being estimated. With mean-length control but without maximum-length control, the broad-family effect on length balance is 6.6 [5.6, 7.7], and *adversative-contrast* is 7.3 [5.0, 9.6]. The corresponding no-length-controls estimates are nearly identical: 6.7 points for the broad family and 7.1 points for *adversative-contrast*. The length-balance claim is reported under mean-length control.

The obvious objection is the right one. Lists and disjunctions are easy places to be parallel: they involve matched daughters, repeated frames, and often explicit co-ordination. The joint-only comparison asks whether *joint-list* and *joint-disjunction* survive against other multinuclear relations. They do, but the effect shrinks to 1.7 points [0.6, 2.8]. The corresponding parison effect is 2.9 [1.0, 4.8], while the isocolon effect is only 0.3 [−0.7, 1.3].

The adversative-only comparison is cleaner. It compares *adversative-contrast* against other adversative relations, where list syntax isn’t the main explanation. Contrast remains positive at 3.0 [1.4, 4.7],

again led by parison. The joint-only baseline consists of *joint-sequence* and *joint-other*; that baseline remains multinuclear and partly co-ordinative.

Two data-scope checks give similar results. In the GUM-only subset, the broad-family effect is 4.0 [3.4, 4.6], the within-joint list/disjunction effect is 1.7 [0.6, 2.9], and the within-adversative contrast effect is 3.4 [1.6, 5.1]. Excluding the 1,963 adjacent rows with discontinuous token intervals leaves the broad-family effect at 4.5 [3.8, 5.1], the within-joint list/disjunction effect at 1.8 [0.6, 3.0], and the within-adversative contrast effect at 3.3 [1.4, 5.2]. A leave-one-genre check also stays stable: the broad-family estimate ranges from 4.0 to 4.3, the within-joint estimate from 1.5 to 2.0, and the within-adversative contrast estimate from 2.8 to 3.2.

The stratified permutation check shuffled target labels 500 times within strata defined by document, explicitness, same-sentence status, colon punctuation, semicolon punctuation, and quartile of mean word length. The model specification was otherwise the same as in the main composite analysis.

This partial null doesn't match local phrase type or root dependency category. It asks whether the result follows from target-label allocation across documents, lengths, punctuation, and explicitness. The observed broad-family effect, the list/disjunction effect, and the adversative-contrast effect all fall above the 97.5th percentile of their shuffled null distributions (Table 8). In 500 stratified shuffles, none of the shuffled label assignments produced an effect as large as the observed one. Because 500 shuffles give only about 0.002 resolution in the tail, I treat this as a calibration check rather than as a precise *p* value. The point isn't family-wise significance testing; it's that the observed effects sit well outside the label-allocation null.

The composite result stays stable across the hand-set weights. Across 66 grid-based nonnegative weightings of length, syntax, and lexical echo, the broad-family effect has a positive interval in every specification. For *adversative-contrast*, 65 of 66 grid specifications are positive and 59 have positive intervals.

Check	Target/outcome	Estimate (points)	Clustered 95% interval
Main	Broad formal balance	4.1	[3.6, 4.7]
Main	<i>joint-list</i> formal balance	4.4	[3.7, 5.1]
Main	<i>adversative-contrast</i> formal balance	2.4	[1.3, 3.5]
Mean-length only	Broad isocolon	6.6	[5.6, 7.7]
Mean-length only	<i>adversative-contrast</i> isocolon	7.3	[5.0, 9.6]
Joint-only	List/disjunction formal balance	1.7	[0.6, 2.8]
Joint-only	List/disjunction parison	2.9	[1.0, 4.8]
Adversative-only	Contrast formal balance	3.0	[1.4, 4.7]
Adversative-only	Contrast parison	5.9	[3.0, 8.8]
GUM-only	Broad formal balance	4.0	[3.4, 4.6]
Continuous spans only	Broad formal balance	4.5	[3.8, 5.1]

Table 7: Selected robustness checks, reported as score-point differences on a 0–100 scale. The joint-only rows compare *joint-list* and *joint-disjunction* against other *joint*-* relations. Mean-length-only rows control log mean length and omit log maximum length. The adversative-only rows compare *adversative-contrast* against other adversative relations. The continuous-spans-only row excludes adjacent rows where either discourse unit has a comma-separated token interval.

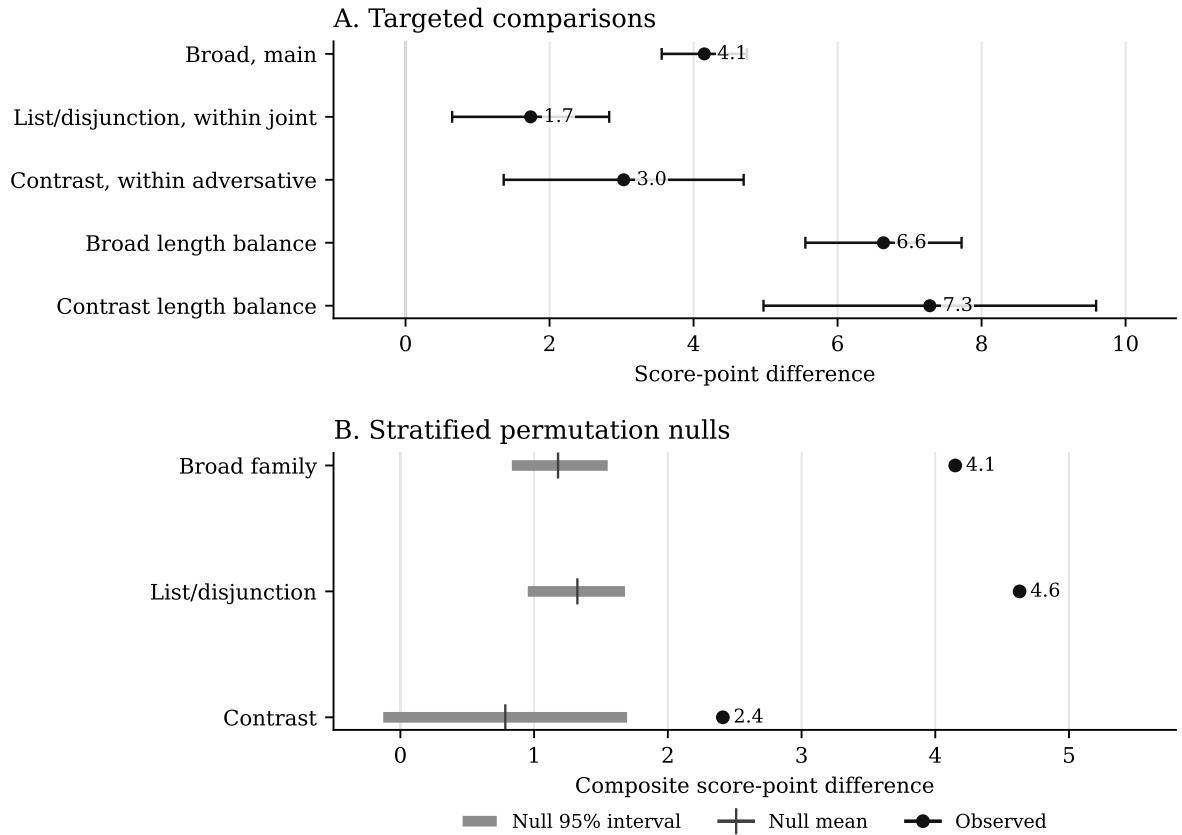


Figure 2: Targeted robustness checks and stratified permutation nulls, reported as score-point differences on a 0–100 scale. Panel A shows adjusted differences with document-clustered 95% intervals. Panel B compares observed composite effects with the null mean and central 95% interval from stratified target-label shuffles.

Target	Observed	Null mean	Null 95%	Strata	Rows	Target rows	<i>p</i>
Broad family	4.1	1.2	[0.8, 1.6]	1,240	8,277	2,174	0.002
List/disjunction	4.6	1.3	[1.0, 1.7]	1,033	7,047	1,659	0.002
<i>adversative-contrast</i>	2.4	0.8	[-0.1, 1.7]	201	1,472	220	0.002

Table 8: Stratified permutation nulls for the composite formal-balance effect, reported as score-point differences on a 0–100 scale. Target labels are shuffled within document, explicitness, same-sentence status, colon/semicolon status, and mean-length quartile strata. The strata, rows, and target-rows columns report the portions of the data that could actually move under the stratified shuffle. The *p* values are two-sided empirical values from 500 permutations, with the standard plus-one correction, and should be read at that resolution.

7 INTERPRETATION

The findings support a restricted version of the rhetorical-function claim. I treat balanced form as a selective way of presenting units as comparable. It's optional even when comparison is present. That view predicts an uneven distribution: balance should cluster where matched presentation has rhetorical value, while remaining absent in many ordinary cases. The corpus result fits that version, but unevenly. Lists and disjunctions supply the largest broad-family effects; adversative contrast supplies the cleaner non-enumerative case.

COMPARABLE SET works here as a descriptive cover term for co-membership relations, including bare co-enumeration. The term allows list structure without treating every list as comparison for its own sake. The cleanest non-enumerative signal in the analysis comes from *adversative-contrast*.

The component results then refine the claim. The dominant effect looks parison-like more than isocolonic in the narrow length-balance sense. The evidence weakens for labels that look rhetorically relevant in the taxonomy but don't necessarily require balanced presentation in the text.

In particular, *adversative-antithesis* as an eRST relation label only partly overlaps with antithesis in the classical theory of figures. The label *adversative-antithesis* keeps the name while losing the balance.² The conservative conclusion leaves antithesis itself intact: the corpus label *adversative-antithesis* serves as a poor direct proxy for the classical figure.

That point remains a limitation as well as a finding. The 80-pair audit contains only eight rows with the *adversative-antithesis* label. These rows support the label-mismatch interpretation, but they're too few to separate label mismatch from a second possibility: genuine semantic antitheses may be expressed through opposition instead of lexical overlap or surface parallelism.

The subset test is defined but not yet reported here. The follow-up worksheet covers all 343 eRST *adversative-antithesis* rows and asks for three judgments: classical antithesis, semantic opposition, and parallel opposition. Once those fields are coded, the analysis compares yes and no rows on formal balance, isocolon, parison, and lexical echo. Until then, the near-zero *adversative-antithesis* result is evidence about the corpus label, not a settled test of classical antithesis itself.

The composite formal-balance score tracks a broader act of placing units into a shared frame. That frame can be contrastive, but it can also be co-ordinative, disjunctive, or list-like. In the latter cases, co-ordination itself favours matched daughters and weakens the rhetorical interpretation. Conversely, some adversative relations have little balance because their function is concession, qualification, or discourse management instead of matched presentation.

Corpus data record ordinary production, not an ideal of trained rhetorical practice. The evidence shows where balance appears in this corpus. It doesn't show where balanced form could improve a text. The language scope is also narrow. These are English corpus data, mostly from GUM, with English word order, English morphology, English punctuation, and English written and spoken segmentation. Nothing here licenses a claim about how isocolon or parison behave in Ancient Greek, Modern Japanese, or any other language whose prosody, morphology, discourse-unit segmentation, or rhetorical training traditions may differ.

The co-ordination caveat matters. Lists and disjunctions are places where parallel syntax is grammatically cheap, so the broad effect can't be read as a pure rhetorical choice. The joint-only robustness check shrinks the effect, which is what that objection predicts. The effect remains positive, and

²The sentence is itself a small ISOCOLON and PARISON, and it asserts an ANTITHESIS. Either the author is demonstrating the apparatus or the apparatus is demonstrating the author.

adversative-contrast remains positive inside the adversative family and under a stratified permutation null.

The safer interpretation says that comparable-set relations attract formal balance beyond genre, document, length, and punctuation differences, while much of that balance follows the syntax of co-ordination. Relation labels don't guarantee classical figures, and classical figures needn't appear where relation labels promise them.

The pattern also clarifies the scope of the corpus study. The evidence stays textual and distributional. It concerns where balanced form occurs, not how readers experience those pairs. A high score by itself still falls short of a rhetorical-figure label. The analysis shows that a graded measure of formal balance is systematically higher in particular discourse-relation contexts. The human audit illustrates why many high-scoring cases are rhetorically relevant and why this remains pilot evidence, pending an independently validated annotation layer.

The RST Signalling Corpus isn't part of the analysis yet because access is still pending. Once available, that corpus can add a later triangulation point.

Work on discourse signalling has shown that coherence relations are often signalled by more than discourse markers, including lexical, syntactic, graphical, semantic, and genre-based cues. That point is developed in the original signalling annotation work and in later studies of RST signals and multiple signals (Das & Taboada, 2018a, 2018b, 2019; Taboada & Das, 2013). This literature gives the project a stronger comparative baseline. If formal balance behaves like a discourse signal, then it should be testable alongside other signal types as well as an isolated figure score.

8 CONCLUSION

A corpus study can't prove that isocolon makes readers perceive comparison as more forceful, and it leaves rhetorical pedagogy untested. It can test a cleaner textual claim: writers use formal balance unevenly across discourse relations, and the relations most associated with comparable-set construals should show more of it. These results support that claim for a broad family of comparison-like and co-ordinative relations, with the largest effects in lists and disjunctions and the cleaner non-enumerative effect in adversative contrast.

The analysis gives selective support to the inherited figure label ANTITHESIS. The narrow adversative effect stays small, and antithesis as a discourse-relation label sits close to zero. That finding sharpens the mapping between relation labels and classical figures. The strongest signal comes from parison-like parallelism; isocolon narrowly understood as length balance contributes less to the main composite effect. Balanced form remains a rhetorical resource, and the corpus evidence supports a claim about comparable-set construals and formal balance more securely than a claim about equal cola alone.

The robustness checks sharpen that conclusion. Length balance appears under mean-length conditioning without maximum length, while parison leads the main composite model. The joint-only comparison also shows that a large share of the broad effect reflects co-ordination itself. The stratified permutation check places the observed effects above document, length, punctuation, and explicitness-matched nulls. The result maps where figural form begins to do work.

DATA AND CODE AVAILABILITY

The analysis scripts, coding materials, and derived audit summaries are kept in the project repository: <https://github.com/BrettRey/isocolon-and-comparative-construal>. The raw GUM/eRST source files are excluded from the repository; the scripts expect a local corpus checkout under data/raw/gum-erst. The RST Signalling Corpus isn't included yet because access is still pending.

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REFERENCES

- Das, D., & Taboada, M. (2018a). RST signalling corpus: A corpus of signals of coherence relations. *Language Resources and Evaluation*, 52(1), 149–184. <https://doi.org/10.1007/s10579-017-9383-x>
- Das, D., & Taboada, M. (2018b). Signalling of coherence relations in discourse, beyond discourse markers. *Discourse Processes*, 55(8), 743–770. <https://doi.org/10.1080/0163853X.2017.1379327>
- Das, D., & Taboada, M. (2019). Multiple signals of coherence relations. *Discours*, 24. <https://doi.org/10.4000/discours.10032>
- Dubremetz, M., & Nivre, J. (2018). Rhetorical figure detection: Chiasmus, epanaphora, epiphora. *Frontiers in Digital Humanities*, 5, 10. <https://doi.org/10.3389/fdigh.2018.00010>
- Fahnestock, J. (1999). *Rhetorical figures in science*. Oxford University Press. <https://doi.org/10.1093/os0/9780195117509.001.0001>
- Fahnestock, J. (2011). *Rhetorical style: The uses of language in persuasion*. Oxford University Press. <https://doi.org/10.1093/acprof:os0/9780199764129.001.0001>
- Harris, R. A., & Di Marco, C. (2017). Rhetorical figures, arguments, computation. *Argument & Computation*, 8(3), 211–231. <https://doi.org/10.3233/AAC-170030>
- Harris, R. A., Di Marco, C., Ruan, S., & O'Reilly, C. (2018). An annotation scheme for rhetorical figures. *Argument & Computation*, 9(2), 155–175. <https://doi.org/10.3233/AAC-180037>
- Kühn, R., & Mitrović, J. (2024). The Elephant in the Room: Ten challenges of computational detection of rhetorical figures. *Proceedings of the 4th Workshop on Figurative Language Processing (FigLang 2024)*, 45–52. <https://doi.org/10.18653/v1/2024.figlang-1.6>
- Kühn, R., Mitrović, J., & Granitzer, M. (2024). Computational approaches to the detection of lesser-known rhetorical figures: A systematic survey and research challenges. <https://doi.org/10.48550/arXiv.2406.16674>
- Lawrence, J., Visser, J., & Reed, C. (2017). Harnessing rhetorical figures for argument mining: A pilot study in relating figures of speech to argument structure. *Argument & Computation*, 8(3), 289–310. <https://doi.org/10.3233/AAC-170026>
- Taboada, M., & Das, D. (2013). Annotation upon annotation: Adding signalling information to a corpus of discourse relations. *Dialogue & Discourse*, 4(2), 249–281. <https://doi.org/10.5087/dad.2013.211>
- Zeldes, A. (2017). The GUM corpus: Creating multilayer resources in the classroom. *Language Resources and Evaluation*, 51(3), 581–612. <https://doi.org/10.1007/s10579-016-9343-x>
- Zeldes, A., Aoyama, T., Liu, Y. J., Peng, S., Das, D., & Gessler, L. (2025). eRST: A signaled graph theory of discourse relations and organization. *Computational Linguistics*, 51(1), 23–72. https://doi.org/10.1162/coli_a_00538